FortiGate Firewall Lab: Deployment & Security Guide

Date: October 2023
Environment: GNS3 Virtual Lab
Device: FortiGate VM (Version 7.x)

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1. GNS3 Environment Setup

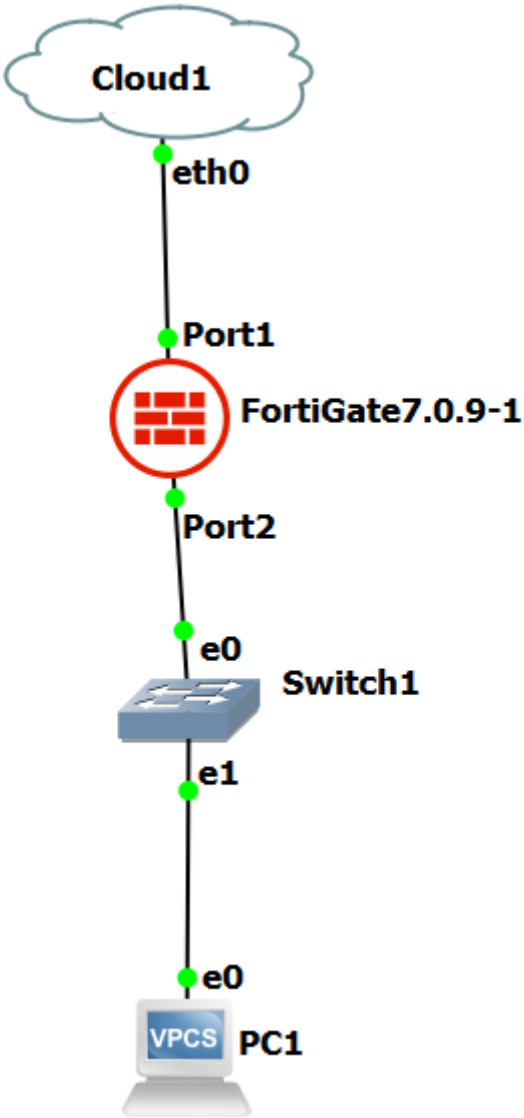
Before booting the firewall, the virtual hardware resources were adjusted to ensure stability and prevent system freezing.

Configuration Steps:

1. **Node:** FortiGate-VM64-KVM
2. **RAM:** Increased to **2048 MB** (Critical for smooth operation).
3. **vCPUs:** Set to 2.
4. **Cloud Node:** Configured to use "Special Ethernet Interfaces" to bridge the virtual lab to the physical Internet adapter.

Connectivity:

- **Port 1 (WAN):** Connected to Cloud Node (Internet).
- **Port 2 (LAN):** Connected to Ethernet Switch -> VPCS (Virtual PC).



GNS3 Topology showing FortiGate connected to Cloud and Switch

2. Initial Initialization (CLI)

Upon first boot, the firewall has no IP address. We used the Command Line Interface (Console) to enable network access for the Management GUI.

Commands Executed:

```
config system interface
  edit port1
    set mode dhcp
    set allowaccess http https ping ssh
  next
end
```

Verification:
We used the command get system interface physical to confirm Port 1 received an IP address from the external network.

```
FortiGate-VM64-KVM # config system interface
FortiGate-VM64-KVM (interface) #     edit port1
FortiGate-VM64-KVM (port1) #         set mode dhcp
FortiGate-VM64-KVM (port1) #         set allowaccess http https ping ssh
FortiGate-VM64-KVM (port1) #     next
FortiGate-VM64-KVM (interface) # end
```

```
FortiGate-VM64-KVM # get system interface physical
== [onboard]
        ==[port1]
                mode: dhcp
                ip: 172.16.16.163 255.255.248.0
                ipv6: ::/0
                status: up
                speed: 1000Mbps (Duplex: full)
                FEC: none
                FEC_cap: none
```

CLI Console showing Port 1 status and IP address

3. Licensing & GUI Access

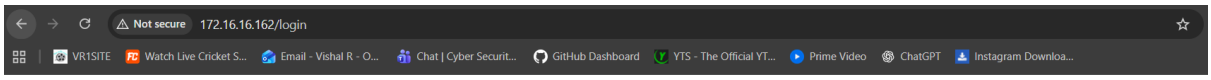
We accessed the FortiGate Web Interface using a web browser pointed to the IP address of Port 1.

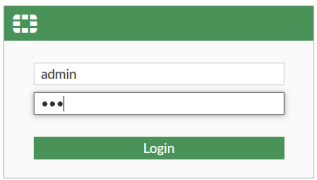
Steps Taken:

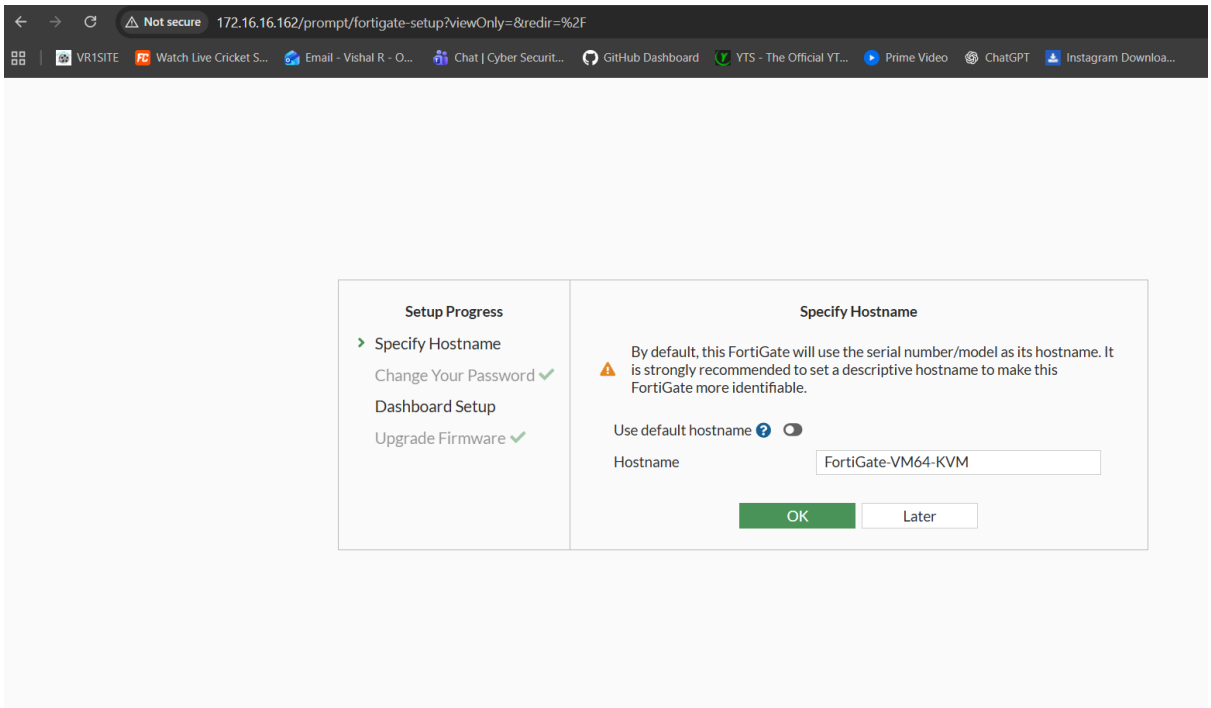
1. **URL:** Accessed http://<Port1_IP> via a standard web browser.
2. **Credentials:** Logged in with default admin (no password), then set a secure password.
3. **Activation:** Selected "Login with FortiCloud" and authenticated with valid Fortinet credentials.
4. **Result:** The VM rebooted, and the permanent evaluation license was activated.

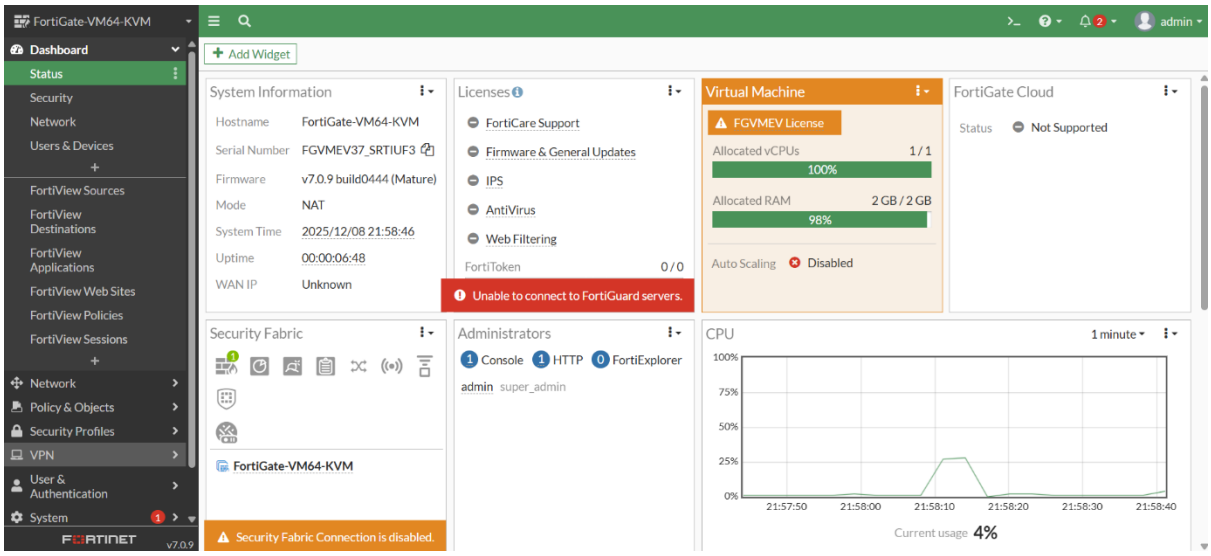
```
FortiGate-VM64-KVM # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=118 time=22.3 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=22.4 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=21.8 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=21.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=21.4 ms

--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 21.4/21.9/22.4 ms
```









FortiGate Dashboard showing System Status and License Valid

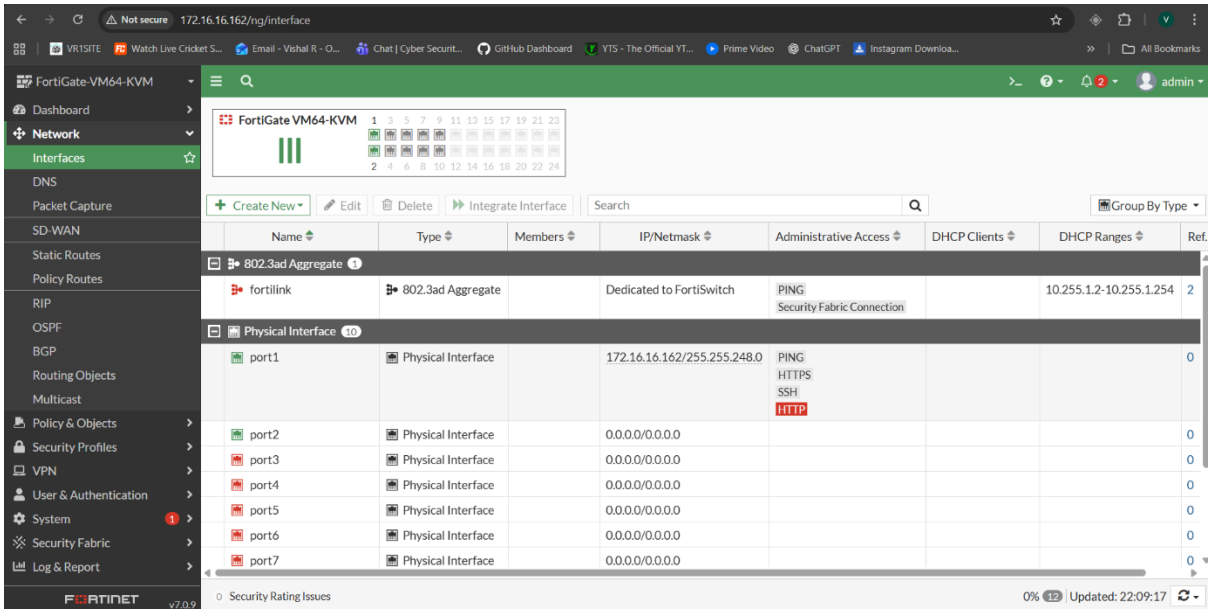
4. LAN Network Configuration

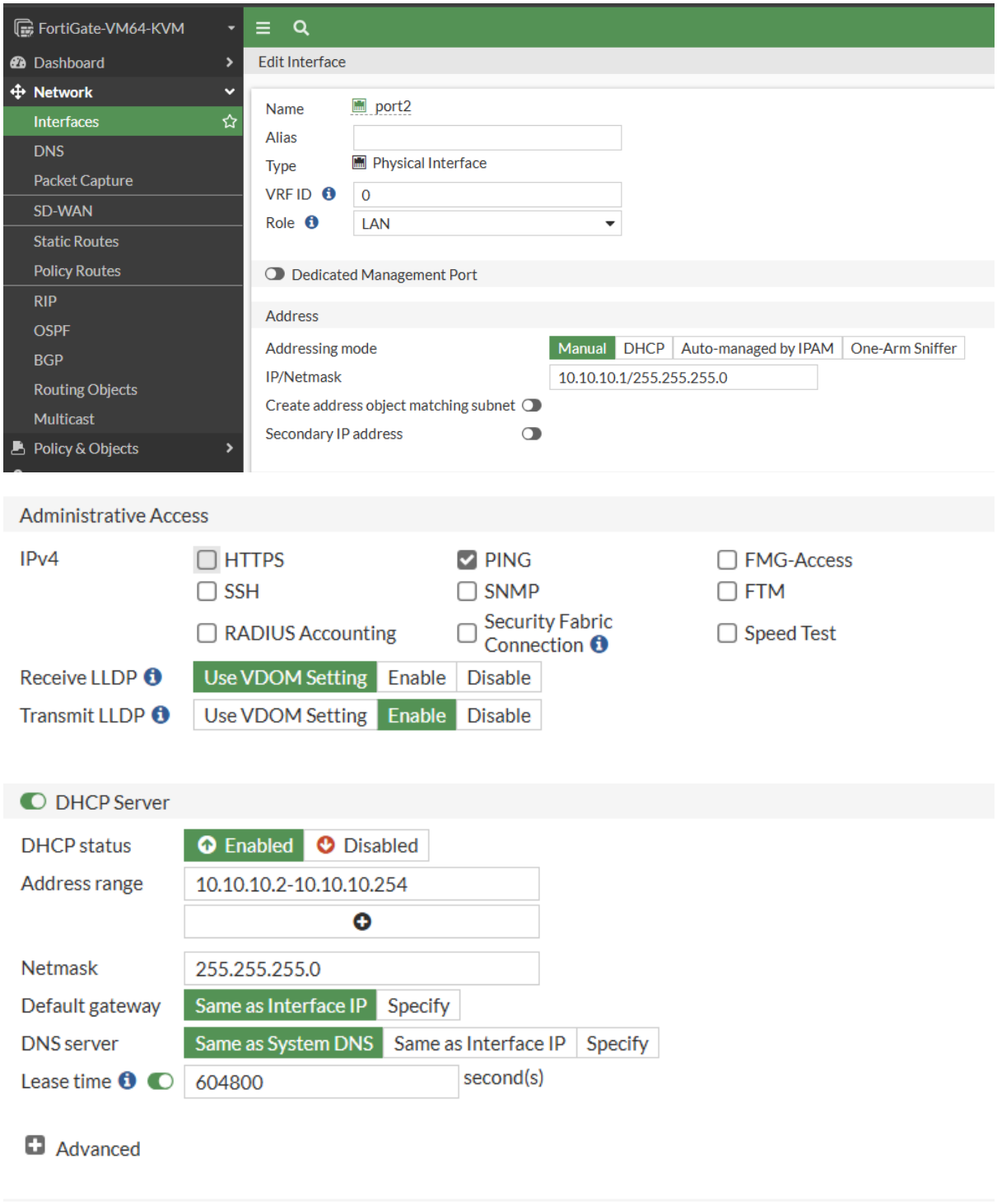
We configured **Port 2** to act as the Gateway for the internal office network.

Settings Applied:

- **Interface:** Port 2
- **Alias:** LAN_OFFICE
- **IP Address:** 10.10.10.1/255.255.255.0
- **Administrative Access:** PING enabled (to allow connectivity testing).
- **DHCP Server: Enabled** (Range: 10.10.10.2 - 10.10.10.254).

Note: We had to save the interface with the IP first, then re-open it to make the DHCP Server options visible.





Network > Interfaces page showing Port 2 configuration and DHCP enabled

5. Client Verification (VPCS)

We verified that the internal network wiring and DHCP services were functional using the Virtual PC.

Tests Performed:

1. **DHCP Request:** Ran ip dhcp in the VPCS console.
    - *Result:* Received IP 10.10.10.2.
2. **Gateway Connectivity:** Ran ping 10.10.10.1.
    - *Result:* Successful reply from the Firewall.

```
PC1> ip dhcp
DDD
Can't find dhcp server

PC1> ip dhcp
DDORA IP 10.10.10.2/24 GW 10.10.10.1

PC1> ping 10.10.10.1

84 bytes from 10.10.10.1 icmp_seq=1 ttl=255 time=0.726 ms
84 bytes from 10.10.10.1 icmp_seq=2 ttl=255 time=1.050 ms
84 bytes from 10.10.10.1 icmp_seq=3 ttl=255 time=0.781 ms
84 bytes from 10.10.10.1 icmp_seq=4 ttl=255 time=0.816 ms
84 bytes from 10.10.10.1 icmp_seq=5 ttl=255 time=1.300 ms
```

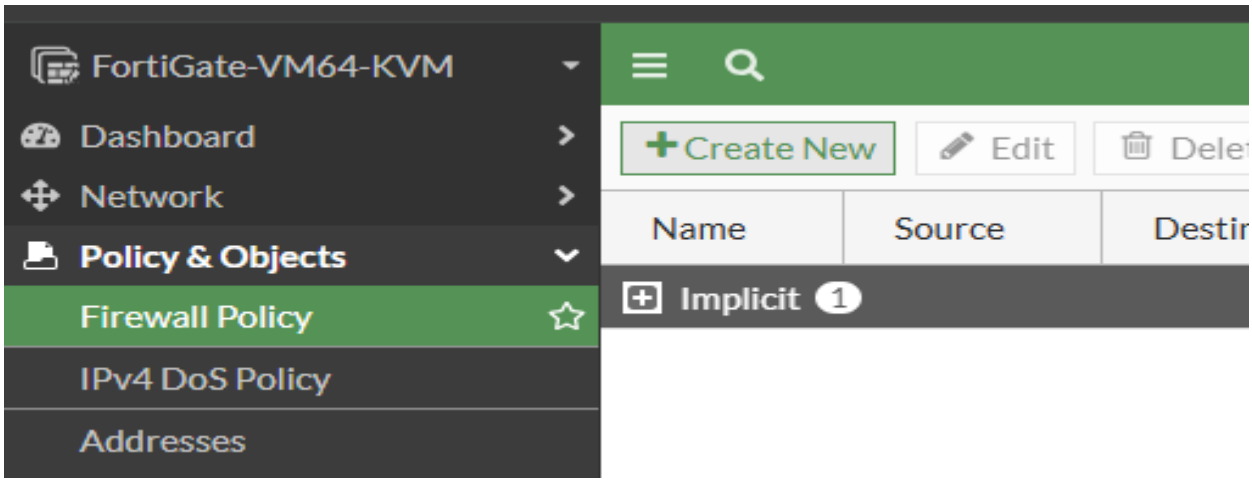
VPCS Console showing successful DHCP DORA and Ping results

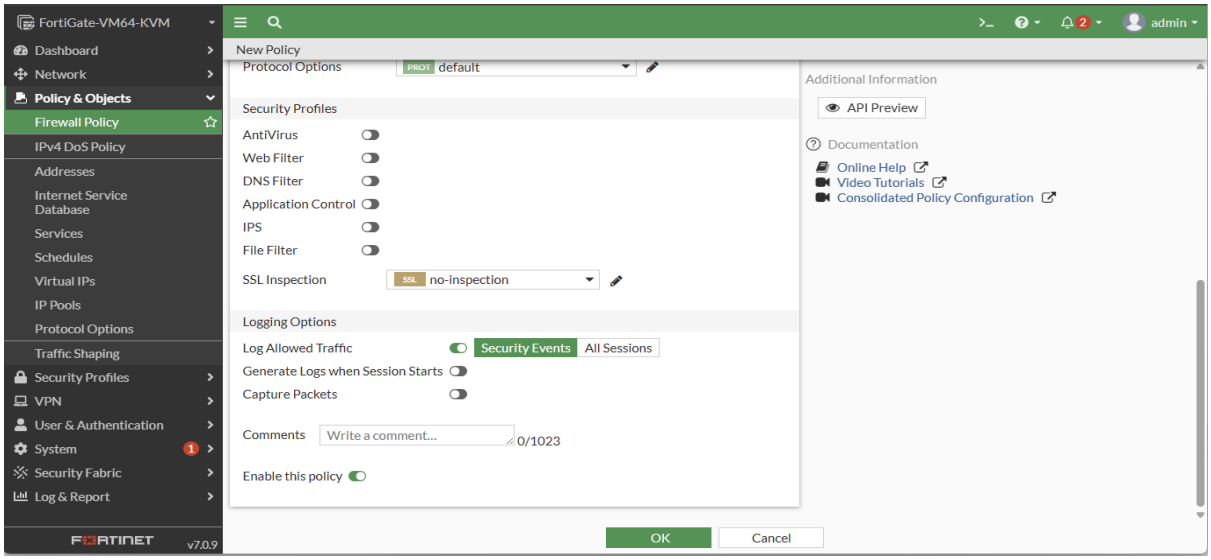
6. Firewall Policy & NAT

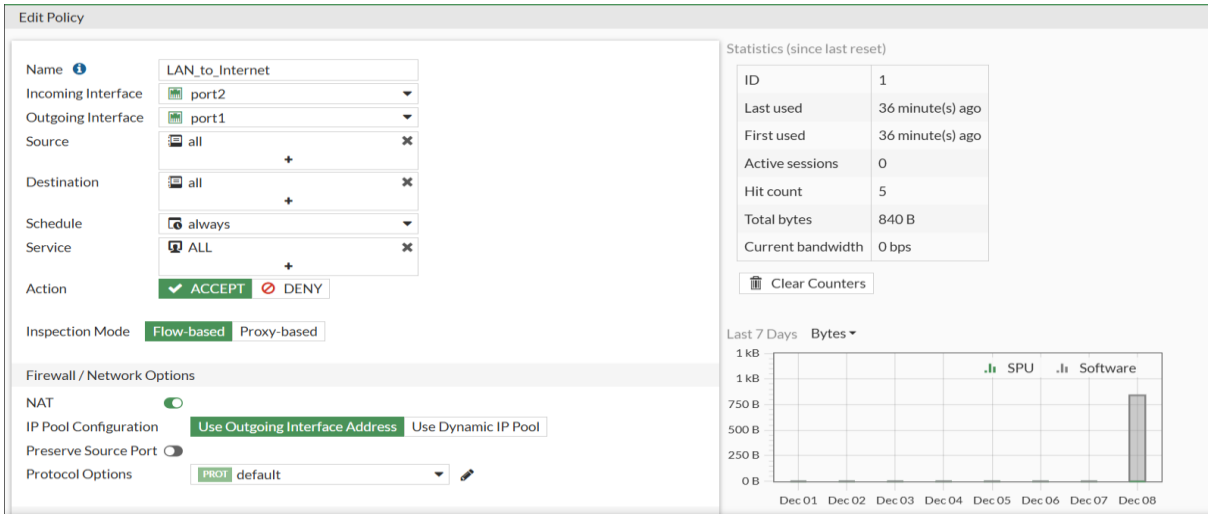
To allow internal users to access the Internet, we created a firewall policy. By default, FortiGate blocks all traffic.

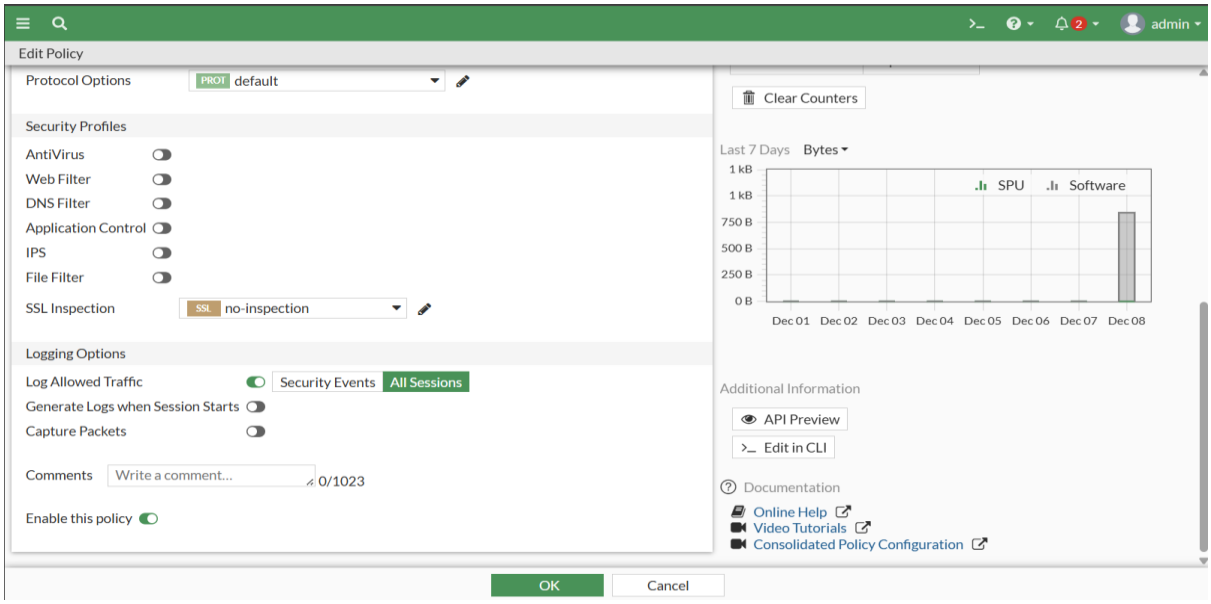
Policy Configuration:

- **Name:** LAN_to_Internet
- **Incoming Interface:** Port 2 (LAN)
- **Outgoing Interface:** Port 1 (WAN)
- **Source/Destination:** ALL
- **Service:** ALL
- **Action:** ACCEPT
- **NAT (Network Address Translation): ENABLED** (Green Toggle).
    - *Why:* This translates the private IP (10.10.10.2) to the public WAN IP so the internet can reply.









Firewall Policy settings showing Source, Destination, and NAT enabled

7. Security Profiles (Web Filtering)

We configured security profiles to inspect traffic and block specific categories.

Configuration:

1. **Profile:** Edited the default Web Filter profile.
2. **Action:** Located the **Gambling** category and set the action to **BLOCK**.
3. **Application:** Applied this Web Filter profile to the LAN_to_Internet firewall policy.
4. **Inspection Mode:** Set SSL Inspection to certificate-inspection.

```
PC1> ping 8.8.8.8

84 bytes from 8.8.8.8 icmp_seq=1 ttl=117 time=22.400 ms
84 bytes from 8.8.8.8 icmp_seq=2 ttl=117 time=22.905 ms
84 bytes from 8.8.8.8 icmp_seq=3 ttl=117 time=23.190 ms
84 bytes from 8.8.8.8 icmp_seq=4 ttl=117 time=23.201 ms
84 bytes from 8.8.8.8 icmp_seq=5 ttl=117 time=22.780 ms
```

Web Filter Profile showing "Gambling" category set to Block

8. Logging & Final Verification

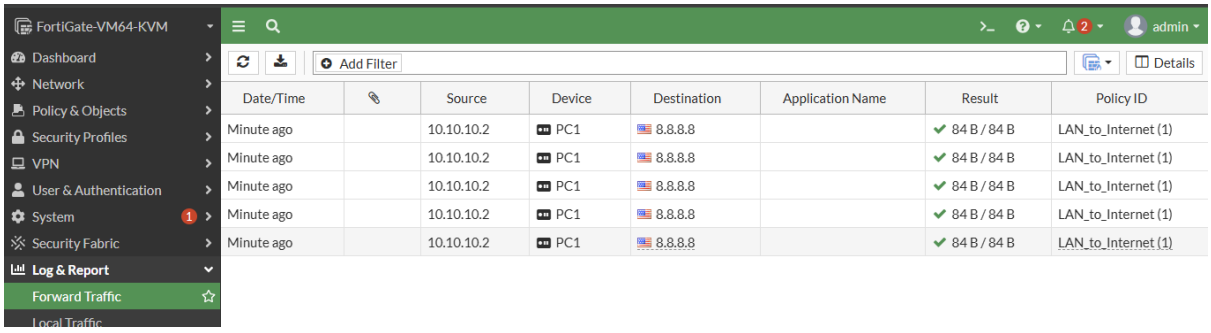
We configured the policy to log all traffic for troubleshooting and audit purposes.

Configuration:

- **Log Setting:** Inside the Firewall Policy, "Log Allowed Traffic" was set to **All Sessions**.

Final Connectivity Test:
From the Virtual PC, we successfully pinged a public DNS server (ping 8.8.8.8).

Log Verification:
We navigated to **Log & Report > Forward Traffic** and confirmed the traffic from 10.10.10.2 to 8.8.8.8 was recorded with an Action of "Accept".



Log & Report > Forward Traffic page showing the Ping logs